

KASTURI B V S V SUHASH

8019807904 ✧ Amaravathi,Guntur

kasturisuhash@gmail.com ✧ [Linkedin](#) ✧ [Github](#)

OBJECTIVE

Electronics engineering student with experience in MATLAB/Simulink and powertrain modeling, seeking an internship in power electronics to apply skills in system simulation and motor control..

EDUCATION

Bachelor of Technology in ECE , VIT-AP University	2022 - Present
CGPA: 8.80	
Intermediate , Sri Chaitanya Junior College	2020-2022
CGPA : 95.2	

PROJECTS

Series Hybrid Electric Vehicle

MATLAB Simulink

[Click Here](#)

- Designed and simulated a Series Hybrid Electric Vehicle (SHEV) using **MATLAB Simulink** and the **Simscape library**, modeling an internal combustion engine (ICE), generator, battery, and motor-drive system.
- Integrated **regenerative braking** and studied key performance metrics such as **battery state of charge (SOC)**, **motor torque**, and **vehicle speed** under different drive conditions.

Underwater Rover

Team Hardware Project

[Click Here](#)

- Designed and implemented an embedded system-based **autonomous underwater vehicle (AUV)** using **Arduino** for motion control and **ESP32** for real-time video transmission and telemetry.
- Developed motor control logic in **C**, utilized **PWM drive circuits** for thruster actuation, and achieved a compact, waterproof hardware layout for operation in confined submerged environments.

Regenerative Braking System

Simulink Simscape

[Click Here](#)

- Developed a regenerative braking system using **MATLAB Simulink**, integrating a bidirectional DC-DC converter with **IGBT switching** for energy recovery and braking control.
- Designed control strategies for torque-based braking, real-time **SOC monitoring**, and energy flow visualization to enhance system efficiency and performance.

SKILLS

Technical Skills	MATLAB,C,Java,perl,Verilog
Development Tools	MATLAB, Simulink, Simscape, PSCAD, Linux, Xilinx Vivado,

EXTRA-CURRICULAR ACTIVITIES

Treasurer, IEEE Student Branch, VIT-AP	2024
Managed financial operations and budgeting for IEEE student activities.Organized AISP-2024, focusing on AI and signal processing.Coordinated the SPAX event, facilitating career development initiatives.	

ACHIEVEMENTS

The Foundational Course in VLSI – NIELIT Chennai

- Gained knowledge of **VLSI fundamentals** and to design using **Verilog** and **FPGA** programming.